

Real 3D simultaneous normal contact: mechanics and exact matrix reduction

Physics-engine research note

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This note supports an executable native 3D benchmark. It does not introduce a novel friction law, claim measured material accuracy, or qualify the failed frictional trajectory references. The implemented optimized profile is frictionless and inelastic. Friction and restitution remain available in the separate pinned Bullet baseline.

1 Three-dimensional bodies and point impulses

Positive action goes from B to A. Lever arms are relative to their respective centres of mass. Use the wedge matrix

$$r^\wedge = \begin{pmatrix} 0 & -r_z & r_y \\ r_z & 0 & -r_x \\ -r_y & r_x & 0 \end{pmatrix}, \quad r^\wedge p = r \times p, \quad v_c = v + (r^\wedge)^\top \omega. \quad (1)$$

With normal impulse magnitude $p_k \geq 0$ and normal $\hat{n}_k$ from B to A,

$$\Delta p_A = \hat{n}_k p_k, \quad \Delta L_A = r_A^\wedge \hat{n}_k p_k, \quad (2)$$

$$\Delta p_B = -\hat{n}_k p_k, \quad \Delta L_B = -r_B^\wedge \hat{n}_k p_k. \quad (3)$$

The world inertia is the full tensor $\mathbb{I}_i = R_i \mathbb{I}_{i,\text{body}} R_i^\top$, not a scalar or a world-fixed diagonal. Authored geometry determines mass, centre and tensor; compound fixtures include all parallel-axis terms. A principal-frame transform supplies Bullet with diagonal body inertia without discarding off-diagonal terms.

2 A many-body contact graph

For a temporary reference length $\ell > 0$, stack each dynamic body's $V_i = (v_i, \ell \omega_i)^\top$ and use

$$H_i = \begin{pmatrix} m_i & 0 \\ 0 & \mathbb{I}_i / \ell^2 \end{pmatrix}, \quad \Delta P_i = \begin{pmatrix} \Delta p_i \\ \Delta L_i / \ell \end{pmatrix}. \quad (4)$$

Scalar square blocks implicitly multiply the appropriate identity. The length only makes the mixed velocity and work pairing convenient; consistent scaling leaves physical results unchanged. Stack dynamic V_i and block-diagonal H_i . The normal contact row for body A is

$$G_{k,A} = (\hat{n}_k^\top \quad \hat{n}_k^\top (r_A^\wedge)^\top / \ell); \quad (5)$$

body B has the negative counterpart. Bodies not participating in a contact have zero blocks; each contact point is a separate row. Prescribed boundaries have zero inverse mobility and enter the relative

normal velocity explicitly:

$$u^- = GV^- + u_{\text{prescribed}}, \quad K = GH^{-1}G^\top, \quad (6)$$

$$\Delta V = H^{-1}G^\top p, \quad u^+ = u^- + Kp. \quad (7)$$

For a prescribed B, its contribution is $-\hat{n}_k^\top(v_B + (r_B^\wedge)^\top\omega_B)$. This is an external actuator; its work must not be omitted.

3 Inelastic unilateral contact and energy

The zero-restitution normal solve is the convex quadratic program

$$\min_{p \geq 0} \frac{1}{2}p^\top Kp + (u^-)^\top p, \quad p \geq 0, \quad u^+ \geq 0, \quad p_k u_k^+ = 0. \quad (8)$$

All incident impulses act through the same matrix, so propagation is simultaneous. With wall work $W_{\text{boundary}} = -p^\top u_{\text{prescribed}}$,

$$\Delta T - W_{\text{boundary}} = p^\top u^- + \frac{1}{2}p^\top Kp \quad (9)$$

$$= -\frac{1}{2}p^\top Kp \leq 0. \quad (10)$$

This frozen-geometry statement is separate from finite-step integration, gravity and position stabilization. Pressure can be nonunique for redundant normals, while the velocity change is unique. The implementation uses Cholesky on positive definite active faces, an active set when bounds bind, and explicit feasibility/complementarity checks. Singular or failed faces disclose fallback; no diagonal compliance or regularization is added.

4 An exact reduction, not ignored friction

The Bullet baseline has one normal and two tangent rows per contact. With zero friction, both tangent impulses are fixed to zero. Partitioning its mobility,

$$\begin{pmatrix} K_{nn} & K_{nt} \\ K_{tn} & K_{tt} \end{pmatrix} \begin{pmatrix} p_n \\ 0 \end{pmatrix} = \begin{pmatrix} K_{nn}p_n \\ K_{tn}p_n \end{pmatrix}. \quad (11)$$

Only the normal unknowns and normal inequalities remain; tangent velocity is allowed to slip. Eliminating fixed-zero rows *before* assembly gives a $c \times c$ scalar mobility instead of $3c \times 3c$: nine times less matrix payload. Eliminating after assembly gives the same physical and algebraic solution but retains unnecessary construction and storage work. The benchmark holds the algorithm, geometry, materials, step guard and integration convention fixed and verifies bitwise-identical state histories. The reduction cannot be applied to nonzero friction, where normal–tangent coupling must remain.

5 The 100 m/s wall test and its scope

An exactly touching row, or packed 3D grid of unit-mass spheres, starts at rest in zero gravity. A prescribed wall/container translates at U , with no friction, restitution or position projection. The inelastic simultaneous solution is

$$v_i^+ = U, \quad \omega_i^+ = 0, \quad x_i(t) = x_i(0) + Ut, \quad W_{\text{boundary}} = N\|U\|^2, \quad T^+ = \frac{1}{2}N\|U\|^2. \quad (12)$$

The fixtures declare initial overlap near 10^{-10} m solely to make floating-point touching-contact discovery stable; it must stay below 10^{-8} m. Contacts use start-of-update wall poses with commanded velocities; walls then advance independently after dynamic integration. This avoids treating one wall travel increment as preexisting penetration. No contents are teleported.

Native collision discovery and orientation are genuinely 3D. The benchmark has 64-body rows along all three axes, a 128-body row, and packed 27/64-body boxes. Every repetition checks all-body trajectories, velocity, spin, penetration, containment, wall work and fallback counts before reporting timing ratios. This ideal normal-contact evidence does not certify chaotic frictional paths, arbitrary initial overlap, all concave swept collisions, experimental materials, or a convergence order for the full engine. Pinned Bullet mechanics and standard active-set QP methods are prior art; the measured change is an implementation and verification result.